

41057 - Heat Exchanger Design Project

Group: Best AU

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Contents

Background	3
Introduction	4
Heat Exchanger Design	5
Laboratory Data Analysis	5
Optimal Case Analysis	6
Modelling	8
Four-Tube Pipe Heat Exchanger	8
Simulation	9
Discussion	11
References	12
Appendices	13
Appendix 1.1 – Lab Result A	13
Appendix 1.2 - Lab Result B	13
Appendix 1.3 - Lab Result C	14
Appendix 1.4 - Lab Result D	15
Appendix 2 – Lab Result Data	16
Appendix 3 – Summarised Results	17
Appendix 4 – 3D Printed Model	19

Background

Currently, it is extremely crucial for all industries to ensure environmental sustainability and operational efficiency. In industrial operations, managing hot wastewater is prioritised for many environmental and operational reasons. There are a lot of hot wastewaters getting produced during various manufacturing processes, which could pose a threat to the environment. Heat exchangers are useful as they help industries reduce heat of wastewater for further cooling purposes and processing.

Heat exchangers typically have multiple inlets and outlets to allow fluid to enter and exit the heat exchangers. They are widely implemented in space heating, refrigeration, air-conditioning, power stations, chemical plants, and sewage treatment.

There are many thermal devices and systems that implement heat exchangers to exchange heat. One example would be internal combustion engines which heats incoming air and cools the coolant by having engine coolant flow through the radiator coils and air flow past the coils.

Counterflow heat exchangers are generally considered to be superior to other types of heat exchangers, such as parallel heat exchangers, because they allow more efficient heat transfer and greater heat exchange rates due to its design of having hot and cold fluids flow in opposite directions, which often leads to energy savings.

Our design offers four tubes in the heat exchanger which is generally better than having one tube because it offers more contact points between the hot and cold fluids due to the greater surface area, thus allowing for greater heat transfer. Additionally, it can enhance the performance of heat exchangers as heat can be transferred at a faster rate with an increased surface area, and pressure drop can be reduced because the flow of fluids is evenly distributed as opposed to a single tube which could lead to higher pressure drop, which often makes heat exchangers costly and inefficient.

Introduction

This project aims to develop an efficient heat exchanger which can provide sustainable cooling of hot wastewater produced from industry processes. The process in utilising the heat exchanger takes safety regulations into account as it is important to conserve energy resources. The focus is to develop an air exchanger that will effectively using a cold-water stream to extract the heat energy from hot wastewater. This process can be employed at various industrial landscapes to adhere to the local regulations when discharging wastewater into sewage systems. The obtained heat energy can be utilised in further stages of the industrial processes, thus conserving more energy.

A laboratory experiment has been conducted for everyone to gain deeper understanding and insights into the designs and the heat exchanger's behaviour. During the experiment, observation of the heat exchanger's behaviour has been recorded as hot wastewater is cooled using a cold-water stream. The heat transfer characteristics and the cooling efficiency have been observed and noted to implement into an improved design of a heat exchanger.

To design a superior model of a heat exchanger, many ideas were considered to enhance the thermal efficiency of the new heat exchanger. A general idea to enhance the efficiency of a heat exchanger is to increase the heat transfer rates, reduce the energy consumption, and make the design more compact and portable.

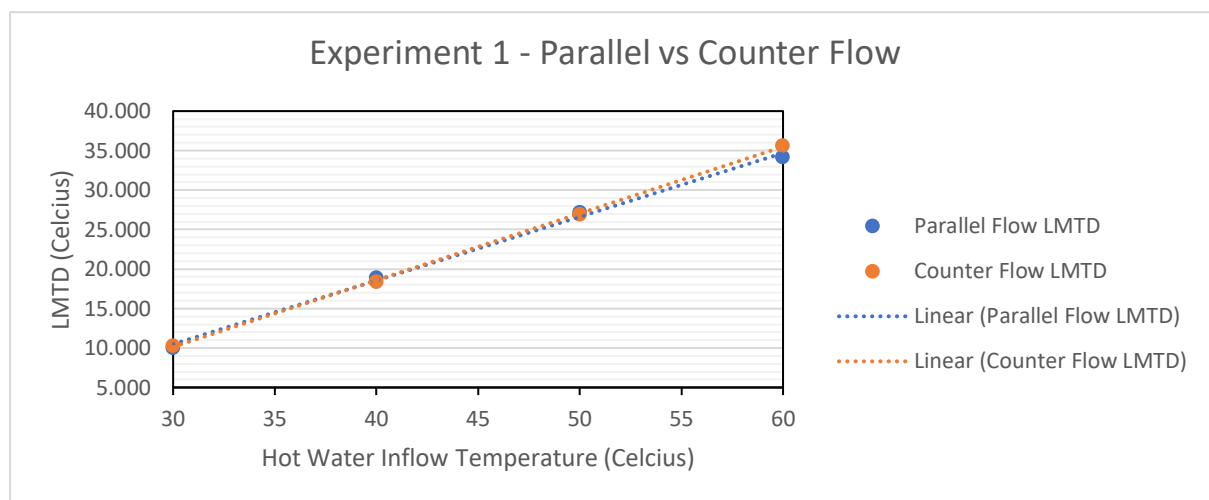
By developing a more efficient design of a heat exchanger, industrial landscapes would benefit tremendously as more efficient heat exchangers could lead to a reduction in environmental impacts and cost savings. This project ultimately aims to create an opportunity to develop a heat exchanger that is cost and energy effective as well as promoting a more sustainable future for industry settings.

Heat Exchanger Design

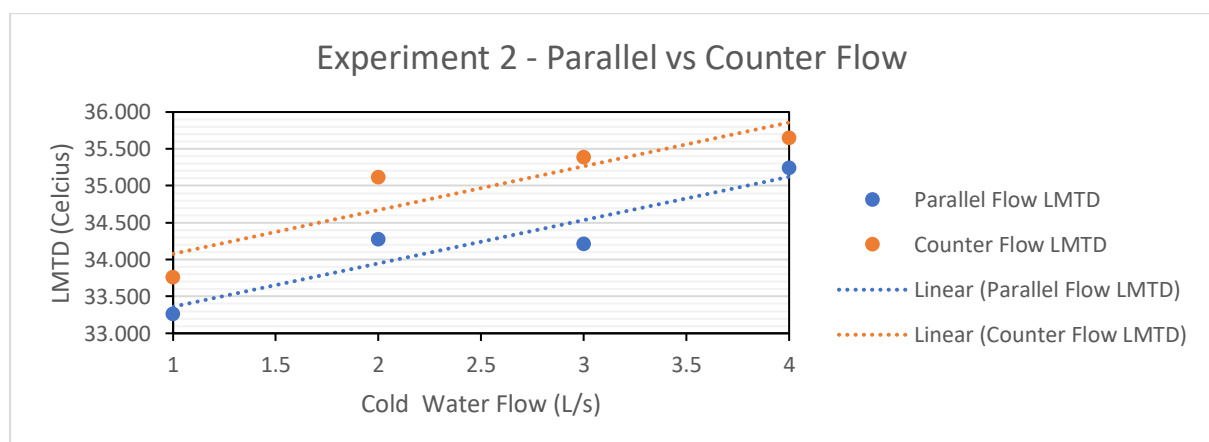
As our design of the four-pipe heat exchanger systems save more heat energy transferred from wastewater to clean water by more efficient cooling. Hot wastewater will flow through four pipes and the heat will be transferred to the cold clean water in the shell. The design is the most appropriate solution of space and heat transfer via the combination of four pipes in one shell of the heat exchanger systems.

Laboratory Data Analysis

Experiment one has the independent variable of the temperature of the hot flow, the dependent variable of the temperature of heat exchanger system and controlled variables of the same liquid, materials, hot and cold flow rate, and room temperature. The resulting data shows there is a correlation of a higher temperature difference as the temperature of the hot water increases with not much change between parallel and counter flow methods.



Experiment two is the change of the cold-water flow rate in the heat exchanger system, so it has independent variable of the cold flow rate, dependent variable of temperature of heat exchanger and, controlled variables of same liquid, hot-water flow rate through the system, materials, and room temperature. The resulting data shows that the increase in cold water flow rate ultimately results in a higher overall temperature difference.



Optimal Case Analysis

The case chosen to analyse was the 60 degrees Celsius hot water, 4L/min cold water flow, and counter flow method. This case was analysed as the resulting data from the experiment proved this to be the most effective case. Based on the data collected and referenced by Çengel et al. (2022) such as density, specific heat, dynamic viscosity, thermal conductivity, and Prandtl number, an effective analysis case was able to be established.

1. The overall heat transfer between the hot and cold water (Q) was calculated through the equation:

$$Q = mc\Delta T$$

- a. m = mass flow rate of the water (Density x Volume Flow Rate) (kg)
 - b. c = specific heat of water interpolated (J/kg*K)
 - c. ΔT = change in temperature (K)
 - d. Q averaged between hot and cold water
2. Next, Reynolds number is calculated for the flowing hot and cold water through the equation:

$$Re = \frac{\rho u D}{\mu}$$

- a. ρ = density of the fluid (kg/m³)
 - b. u = speed of the fluid (Flow Rate/ Cross Sectional Area) (m/s)
 - c. D = hydraulic diameter of the fluid ($\frac{4 \cdot A}{\Sigma P}$) (m)
 - d. μ = dynamic viscosity interpolated (kg/m*s)
3. Using the Reynolds number, Colburn's Equation was used to calculate Nu:

$$Nu = 0.023 Re^{0.8} Pr^{1/3}$$

- a. Pr = Prandtl number interpolation
4. Using Nu, the convection heat coefficient is then calculated:

$$h = \frac{Nu \cdot k}{D}$$

- a. k = thermal conductivity interpolated (W/K*m)
 - b. D = hydraulic diameter (m)
5. Finally, the total resistance value was compared to find the length of the equivalent heat exchanger

$$R_{Total} = \frac{T_H - T_C}{Q} = \frac{1}{\pi D_H h_H L} + \frac{\ln \left(\frac{R_O}{R_I} \right)}{2\pi k L} + \frac{1}{\pi D_C h_C L}$$

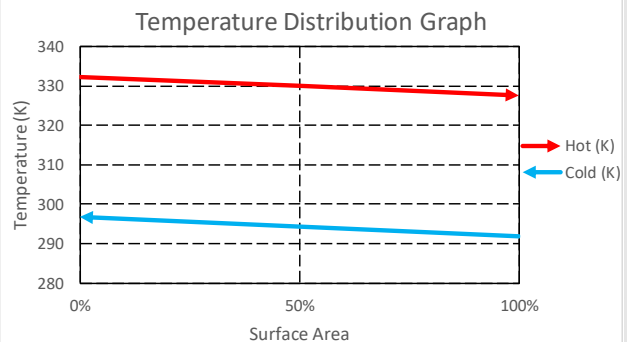
6. To calculate the length, the goal-seek function was used to make both R totals equal. This is further shown through the below spreadsheet.

Experiment 2					
Saturated Water Properties (From References)					
Temp (Celcuis)	Density	Specific Heat (J/kg·K)	Dynamic Viscosity (*10^-3) (kg/m*s)	Thermal Conductivity (W/K*m)	Prandtl Number
20	998	4182	1.002	0.598	7.01
25	997	4180	0.891	0.607	6.14
55	985.2	4183	0.504	0.649	3.25
60	983.3	4185	0.467	0.654	2.99
Intrapolated					
21.15	997.77	4181.54	0.97647	0.60007	6.8099
56.7875	984.52075	4183.715	0.4907725	0.6507875	3.15705

Measured Temperatures			
Area	0%	50%	100%
Hot (Celsius)	59.2	57	54.375
Hot (K)	332.35	330.15	327.525
Cold (Celsius)	23.6	21.15	18.7
Cold (K)	296.75	294.3	291.85

Calculations

	Hot	Cold	Temperature (K) Temperature (K) 340 330 320 310 300 290 280 0%
Initial Temperature (Celcius)	56.7875	21.15	
Flow	Counter		
Flow Rate (L/min)	4	4	
Average Density (kg/m3)	984.521	997.770	
Flow Rate (kg/s)	0.06563	0.06652	
Average Cp (J/kg·K)	4183.71500	4181.54000	
Average C (J/s·K)	274.59695	278.14768	
Q (W)	1324.93028	1362.92362	
Area (sqm)	0.00008	0.00020	
Velocity (m/s)	0.84883	0.33157	
Dynamic Viscosity (kg/m*s)	0.00050	0.00100	
Re	16581	2641	
Pr	3.25	7.01	Inner Tube S. Steel (AISI 302)
Colburn Equation (Nu)	80.92	24.05	
Thermal Conductivity (W/K*m)	0.649	0.598	15.1
h (W/K*m2)	5251.59	1797.67	
Inner Radius (mm)	0	6	5
Outer Radius (mm)	5	10	6
Hydraulic Diameter (m)	0.01	0.008	
Resistance	0.009697934	0.014165431	0.003074689

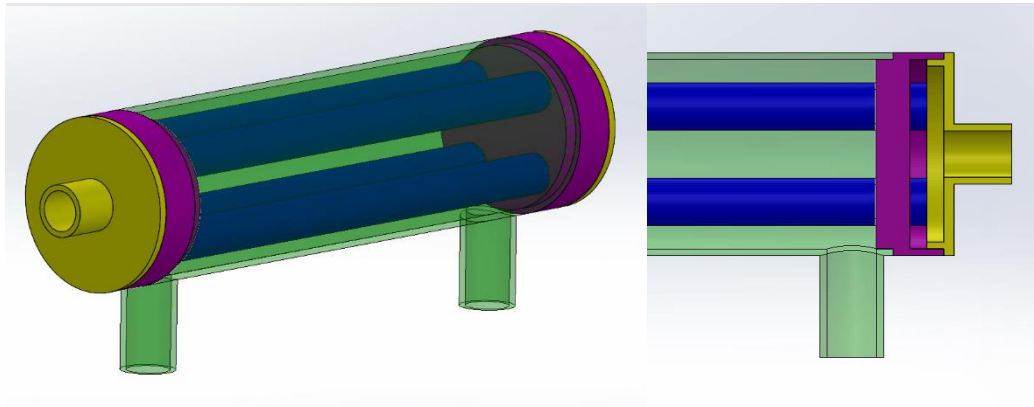


Change in Temperature 1	35.6
Change in Temperature 2	35.675
LMTD	35.637
R(Total) [SUM(R)]	0.0269
R(Total) [TH-TC/Q]	0.0265
Length of Design Tube (m)	0.625

Dependent on Length
USE GOAL SEEK (UNDER WHAT-IF ANALYSIS - DATA) TO FIND
LENGTH (R(TOTAL) = R(TOTAL))

Modelling

Four-Tube Pipe Heat Exchanger



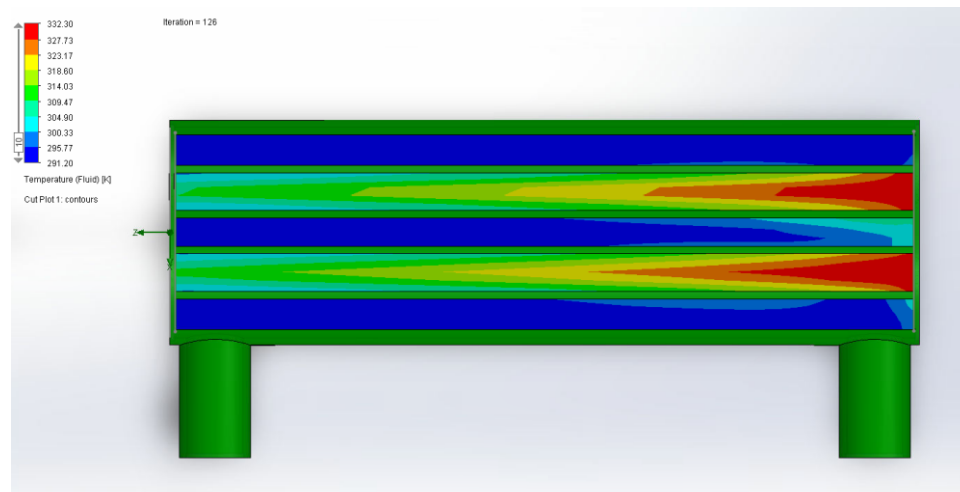
This design is improved with four inner tubes instead of one. The four tubes are joined at both ends by a one-to-four split cap. Cold water inlet and outlet is built into the green shell itself. By increasing the number of tubes, we expect these improvements in the heat exchange performance:

- **Increased Surface Area:** More tubes mean a greater total surface area for heat transfer. The increased surface area allows for more contact between the tube-side fluid and the shell-side fluid, facilitating a more efficient transfer of thermal energy.
- **Improved Turbulence and Mixing:** When multiple tubes are used, the flow of the fluids inside the shell can become more turbulent. Turbulent flow promotes better mixing of the fluids, ensuring that fresh fluid is continuously brought into contact with the tube walls, enhancing heat transfer. With the addition of the end caps, turbulent flow is introduced.
- **Reduced Fouling and Scaling:** In some applications, such as those involving fluids with impurities, fouling and scaling on the tube walls can reduce heat transfer efficiency. Using multiple tubes can mitigate this issue because any fouling or scaling that occurs on one tube may not affect the others, allowing the heat exchanger to continue operating effectively.

Simulation

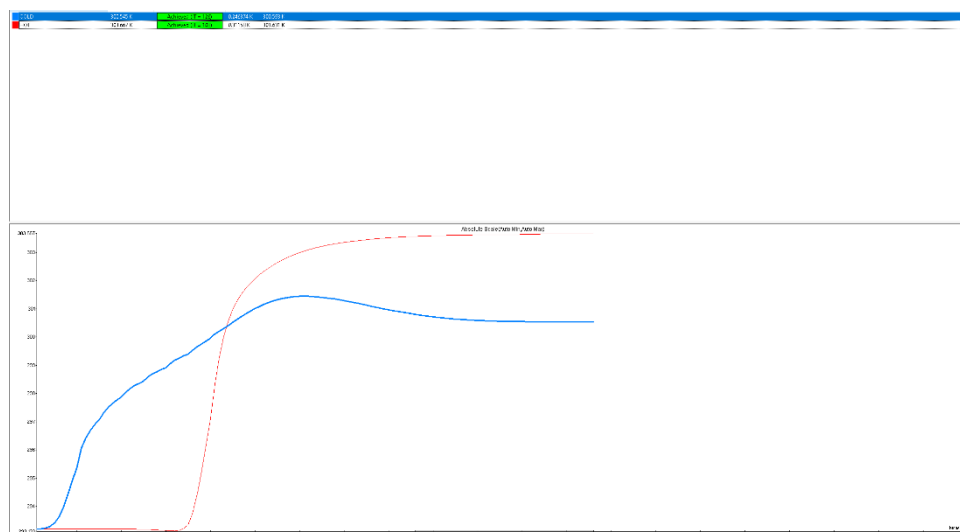
1. FEA Model:

The FEA Model below shows the cross section of the heat transfer as it is completed. Different colours are depicted to reflect upon the temperature of the fluids at different areas of the cross section. As seen in this picture, the hot tubes start at high temperature of 332 Kelvin and exits the outlet at a much colder temperature of 310 Kelvin. On the other hand, the cold tube enters the heat transfer at 291.2 Kelvin and ended at 298 Kelvin. This picture clearly demonstrates the heat transfer in action in a visually sound way.



2. Cold Tube Outlet and Hot Tube Outlet Temperature Over Time:

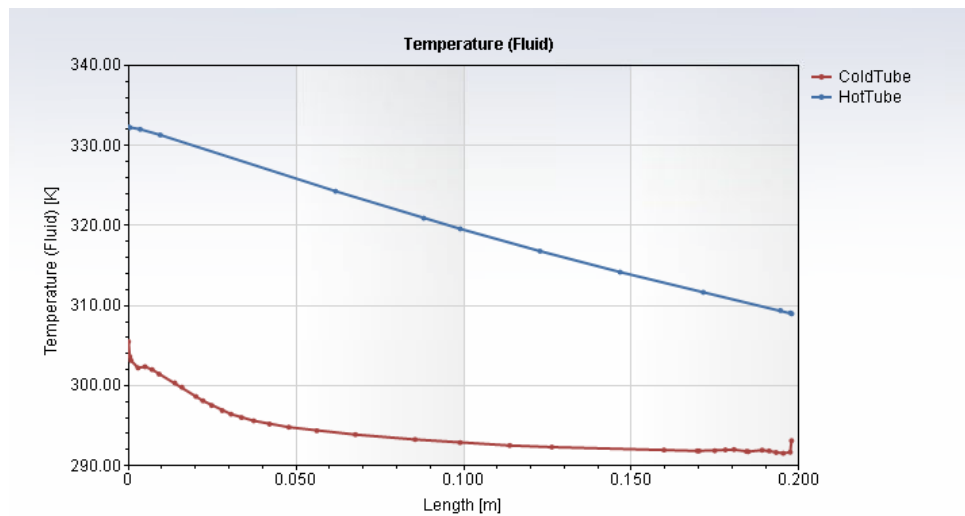
The time graph below depicts the temperature at the outlets of both Cold Tube and Hot Tubes. At time 0, they start at ambient temperature of 293.2 Kelvin, but takes massive turns over time. Cold tube (defined by the blue line) rises in temperature as it travels to the outlet, since it has gained heat from the hot tubes. On the other hand, the hot outlet



3. Cold Tube and Hot Tube Temperature Over Total Length:

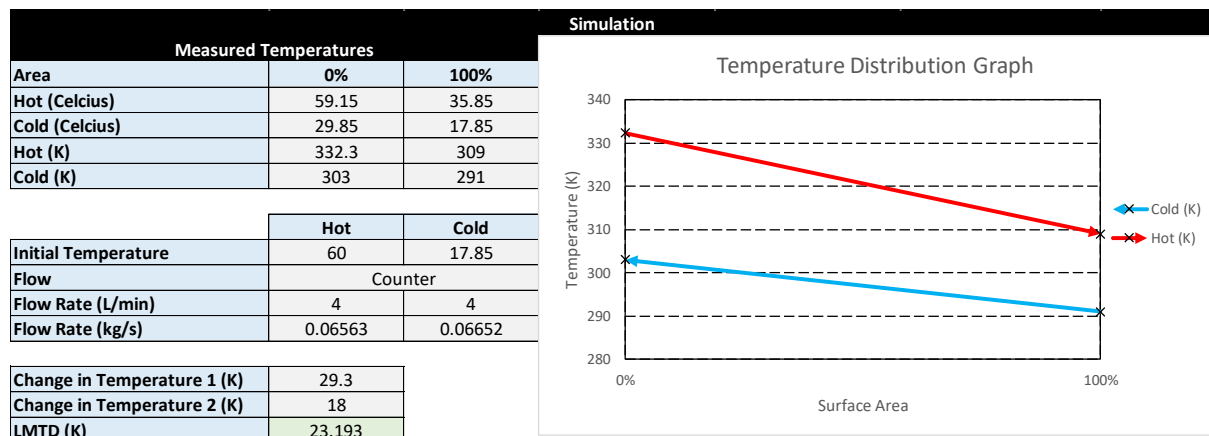
The graph below depicts the temperature of both the Cold Tube and the Hot Tube over the length of them (200mm). This shows a visual representation of the temperature of the fluid at different areas of the tube. For example, the Hot Tube chart shows how its temperature drop from 332 Kelvin from

the entry down to 308 Kelvin at the exit. On the other hand, the Cold Tube chart shows how its temperature rises from 291 Kelvin to 303 Kelvin.



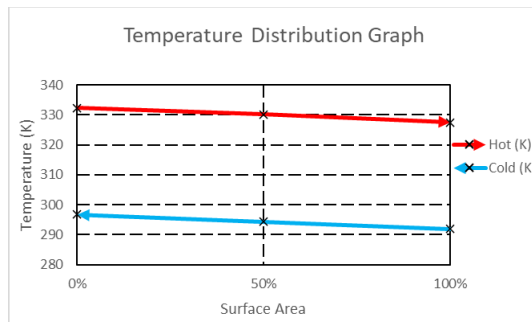
4. Simulation Analysis

Using the input and output temperatures for the hot-water flow and cold-water flow of the simulation, a temperature distribution graph of the following is produced, as well as a log mean temperature difference of 23.2 K. The main noticeable part of the graph includes the steeper slope of the hot water flow which may be due to the large temperature drop when the four hot water tubes connect.



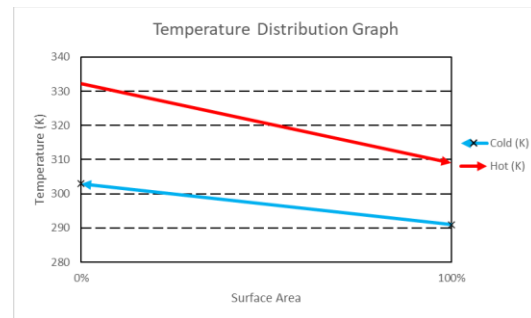
Discussion

Experiment 2



LMTD = 35.6

Simulation



LMTD = 23.2

Comparison

Both experiments have the same initial conditions: same inner tube material, counter flow method, liquid, and inflow temperatures. This indicates that a valid comparison between the results of both experiments can be carried out. By comparing the temperature distribution graphs and log mean temperature differences of the experimental and the simulation, it can be concluded that the simulation produces a more promising heat exchange. This is because the Four-tube Pipe heat exchanger has more surface area, thus allowing greater heat transfer, making the simulation more efficient. Although the LMTD of the simulation is lower than that of the experimental, the simulation ultimately lowers the hot water outflow by a substantial amount.

Since our prototype is made by 3D Printing, it only serves as a look-and-feel model for this submission. However, there is a few improvements that we would like to suggest if this was made for an actual heat exchange purpose.

1. Tube being made of metal:

Metal such as AISI 302 has a much higher conduction coefficient than plastic. This means that the rate at which heat can be transmitted through the body of Steel is much higher than that of plastic. Therefore, it will enhance the heat exchange efficiency by a lot because heat can be transferred from the hot tube fluid to the steel tube, to the cold fluid.

2. Manufacturing:

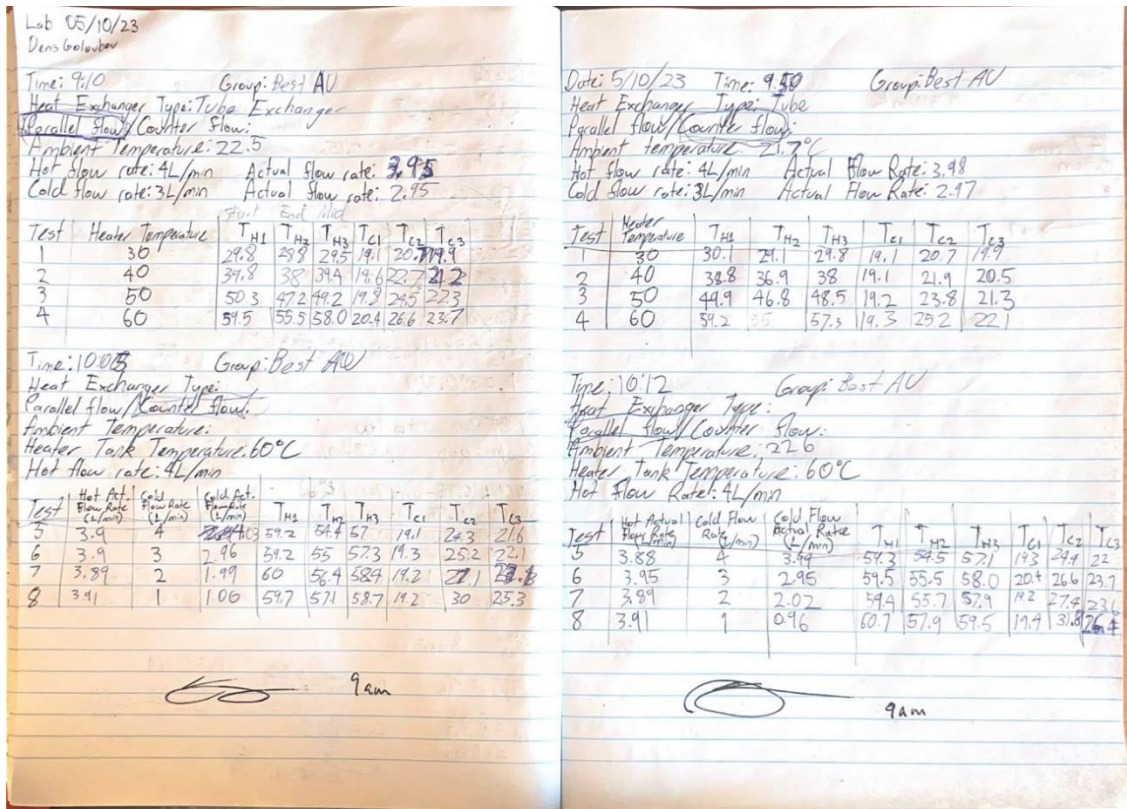
Metal Extrusion production method will be needed to produce the hot fluid tubes. This ensures that production happens at a much higher efficiency, to produce these units in a timely manner. This will decrease production cost as well as increase sales performance. 3D printing is only meant for rapid prototyping purposes and is not intended for actual commercial use.

References

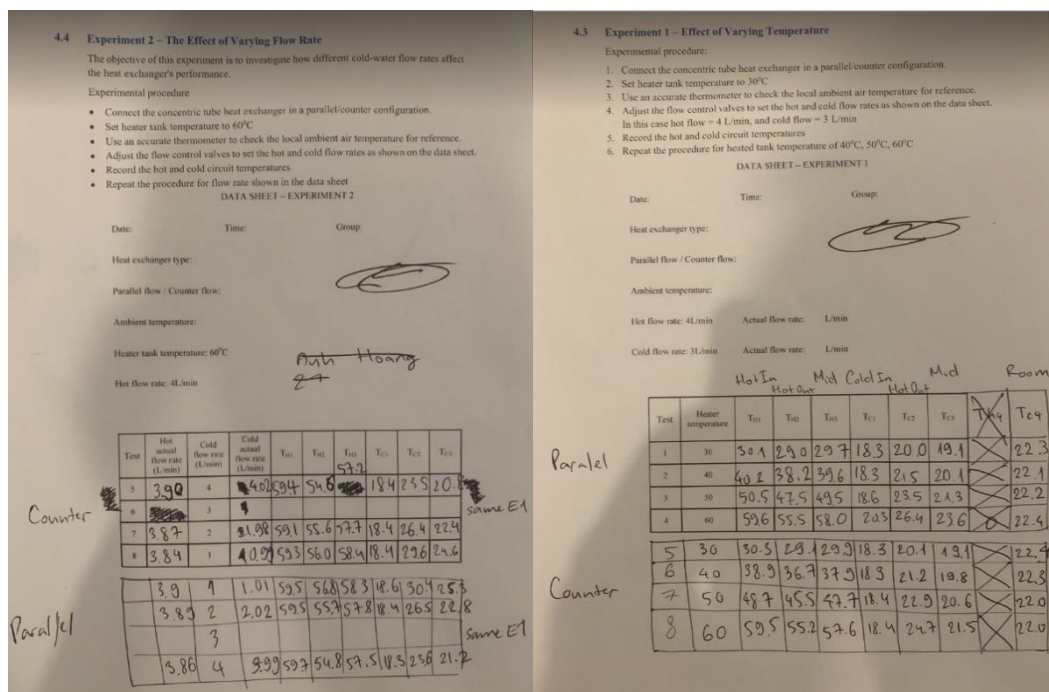
Çengel, Y. A., Cimbala, J. M., & Ghajar, A. J. (Afshin J. (2022). *Fundamentals of thermal-fluid sciences* (Sixth edition.). McGraw Hill.

Appendices

Appendix 1.1 – Lab Result A



Appendix 1.2 - Lab Result B



Appendix 1.3 - Lab Result C

Counter flow									
Ngo Trong Tuan Pham 2459 0396									
Test	Hot actual flow rate (l/min)	Cold flow rate (l/min)	Cold actual flow rate (l/min)	T_{H1}	T_{H2}	T_{H3}	T_{C1}	T_{C2}	T_{C3}
5	3.86	4	4.00	59.0	54.1	56.8	18.2	22.3	20.6
7	3.86	2	2.00	59.8	56.1	58.3	18.3	26.5	27.4
8	3.88	1	1.00	60.2	57.5	59.2	18.4	30.4	24.6

Parallel flow									
Test	Heater Temperature	T_{H1}	T_{H2}	T_{H3}	T_{C1}	T_{C2}	T_{C3}		
1	30	29.7	28.6	29.3	17.9	19.2	18.7		
2	40	40.0	36.0	39.3	17.9	21.0	19.6		
3	50	49.7	46.6	48.6	17.9	22.7	20.6		
4	60	59.7	55.4	58.0	18.6	24.9	22.0		

at ~~60~~ ~~18.2~~
for temperature: 22.3

Purchase
03/10/2023

Counter flow									
Test	Heater	T_{H1}	T_{H2}	T_{H3}	T_{C1}	T_{C2}	T_{C3}		
1	30	29.8	28.8	29.4	18.0	19.7	18.8		
2	40	39.8	37.6	38.9	18.1	21.2	19.6		
3	50	49.6	46.3	48.1	18.2	22.8	20.4		
4	60	59.7	55.1	57.7	18.2	24.5	21.3		

Appendix 1.4 - Lab Result D

4.3 Experiment 1 – Effect of Varying Temperature

Experimental procedure:

1. Connect the concentric tube heat exchanger in a parallel/counter configuration.
2. Set heater tank temperature to 30°C.
3. Use an accurate thermometer to check the local ambient air temperature for reference.
4. Adjust the flow control valves to set the hot and cold flow rates as shown on the data sheet.
In this case hot flow = 4 L/min, and cold flow = 3 L/min
5. Record the hot and cold circuit temperatures
6. Repeat the procedure for heated tank temperature of 40°C, 50°C, 60°C

DATA SHEET – EXPERIMENT 1

Date: Time: Group:

Heat exchanger type: Double Pipe Heat Exchanger 11am

Parallel flow / Counter flow: 5/10/23

Ambient temperature:

Hot flow rate: 4L/min Actual flow rate: 3.99 L/min

Cold flow rate: 3L/min Actual flow rate: 2.96 L/min

Test	Heater temperature	T _{H1}	T _{H2}	T _{H3}	T _{C1}	T _{C2}	T _{C3}	T _{C4}
1	30	29.8	28.8	29.4	18.3	19.9	19.1	22.4
2	40	40.6	38.4	39.6	18.3	21.6	19.9	22.4
3	50	49.5	46.6	48.5	18.5	23.3	21.1	22.3
4	60	59.1	55.0	57.5	20.1	26.2	23.4	22.3°C

↑ ↑ ↑ ↑ ↑
 Hot in out put middle cold in out put middle

4.4 Experiment 2 – The Effect of Varying Flow Rate

The objective of this experiment is to investigate how different cold-water flow rates affect the heat exchanger's performance.

Experimental procedure

- Connect the concentric tube heat exchanger in a parallel/counter configuration.
- Set heater tank temperature to 60°C
- Use an accurate thermometer to check the local ambient air temperature for reference.
- Adjust the flow control valves to set the hot and cold flow rates as shown on the data sheet.
- Record the hot and cold circuit temperatures
- Repeat the procedure for flow rate shown in the data sheet

DATA SHEET – EXPERIMENT 2

Date: Time: Group:

Heat exchanger type: Double Pipe Heat Exchanger

Parallel flow / Counter flow:

Ambient temperature:

Heater tank temperature: 60°C

Hot flow rate: 4L/min

Test	Hot actual flow rate (L/min)	Cold flow rate (L/min)	Cold actual flow rate (L/min)	T _{H1}	T _{H2}	T _{H3}	T _{C1}	T _{C2}	T _{C3}
5	3.89	4	4.00	59.6	54.8	57.9	18.3	27.6	21.2
6		3							
7	3.94	2	2.02	59.6	55.8	57.9	18.4	26.5	22.9
8	3.95	1	1.01	60.9	57.6	59.2	18.6	30.7	25.4

Counter flow

5/10/2023

11:00am

Experiment 1:

Test	Heater temperature	T _{H1}	T _{H2}	T _{H3}	T _{C1}	T _{C2}	T _{C3}	T _{C4}
1	30	30.1	29.0	29.8	18.3	20.0	19.1	22.4
2	40	40.4	38.1	39.4	18.3	21.5	19.8	22.4
3	50	49.5	46.2	48.0	18.4	23.0	20.7	22.1
4	60	59.4	55.1	57.5	18.4	24.7	21.5	22.0

Experiment 2:

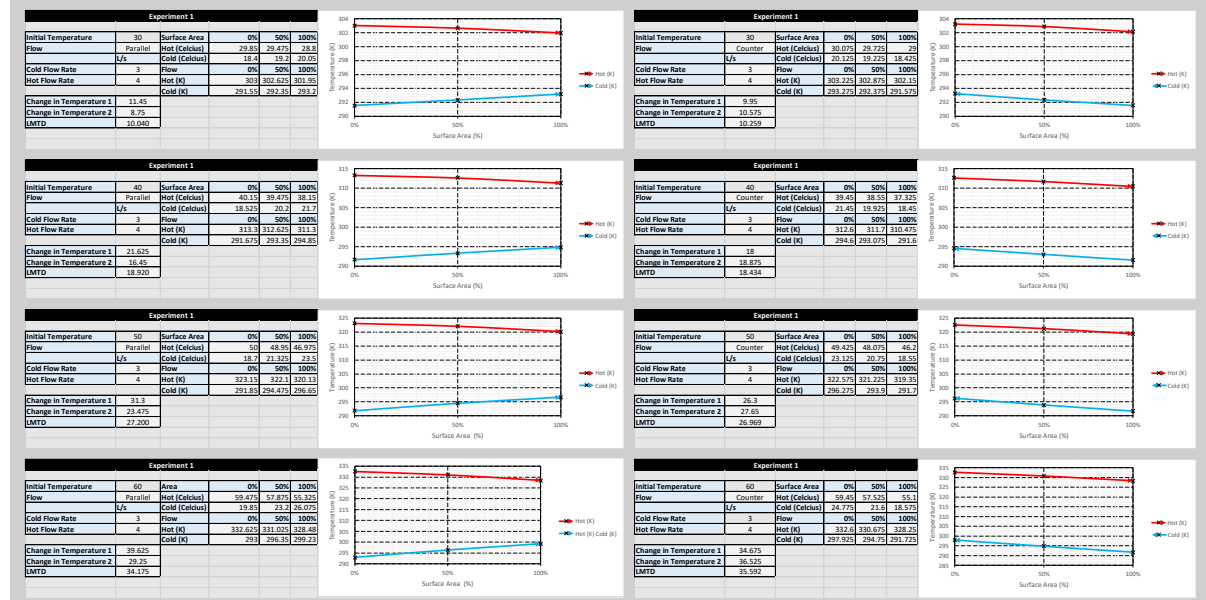
Test	Hot actual flow rate (L/min)	Cold flow rate (L/min)	Cold actual flow rate (L/min)	T _{H1}	T _{H2}	T _{H3}	T _{C1}	T _{C2}	T _{C3}	T _{H4}
5	3.85	4	4.02	59.4	54.5	57.1	18.4	27.6	20.8	0.0
6		3								
7	3.88	2	1.04	59.1	55.5	57.7	18.4	26.4	22.4	0.0
8	3.91	1	0.99	59.2	56.7	58.3	18.5	29.9	24.6	0.0

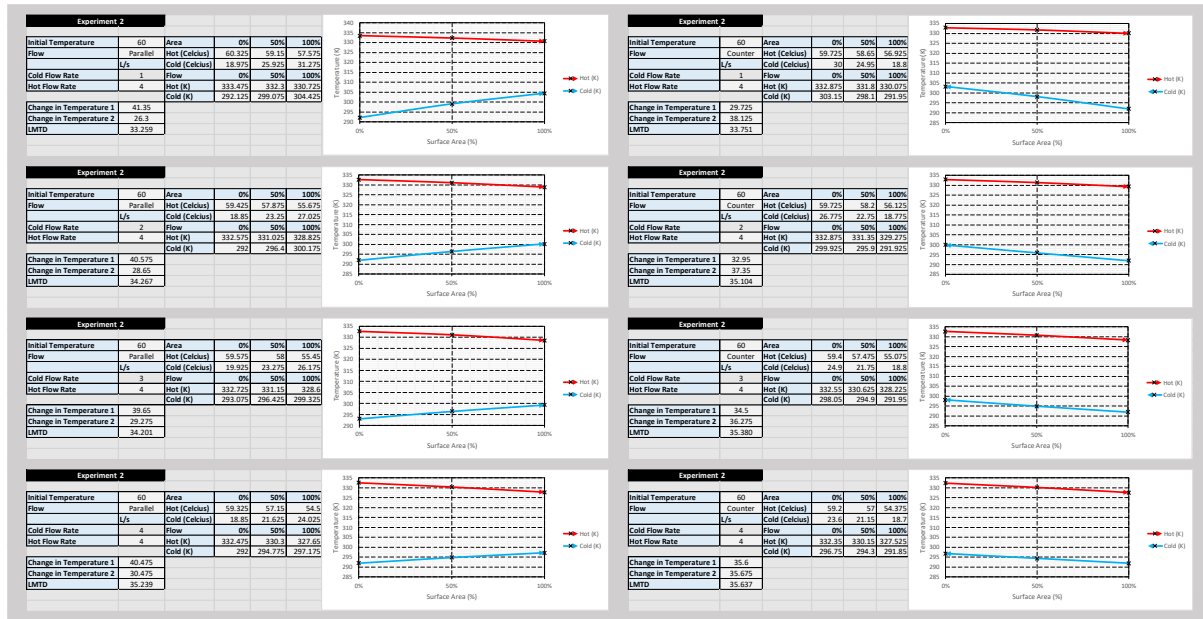
Appendix 2 – Lab Result Data

Responsibility	Experiment Number	Parallel/Counter Flow?	Ambient Temperature	Hot Flow Rate	Actual Hot Flow Rate	Cold Flow Rate	Actual Cold Flow Rate	Heater Temperature	TH1	TH2	TH3	TC1	TC2	TC3	Comments
Denis Goloubev	1	Parallel	22.5	4	3.95	3	2.95	30	29.8	28.8	29.5	19.1	20.7	19.9	
Denis Goloubev	1	Parallel	22.5	4	3.95	3	2.95	40	39.8	38	39.4	19.6	22.7	21.2	
Denis Goloubev	1	Parallel	22.5	4	3.95	3	2.95	50	50.3	47.2	49.2	19.8	24.5	22.3	
Denis Goloubev	1	Parallel	22.5	4	3.95	3	2.95	60	59.5	55.5	58	20.4	26.6	23.7	
Denis Goloubev	1	Counter	21.7	4	3.98	3	2.97	30	30.1	29.1	29.8	19.1	20.7	19.9	
Denis Goloubev	1	Counter	21.7	4	3.98	3	2.97	40	38.8	36.9	38	19.1	21.9	20.5	
Denis Goloubev	1	Counter	21.7	4	3.98	3	2.97	50	49.9	46.8	48.5	19.2	23.8	21.3	
Denis Goloubev	1	Counter	21.7	4	3.98	3	2.97	60	59.2	55	57.3	19.3	25.2	22.1	
Denis Goloubev	2	Counter	22	4	3.9	4	4.03	60	59.2	54.4	57	19.1	24.3	21.6	
Denis Goloubev	2	Counter	22	4	3.9	3	2.96	60	59.2	55	57.3	19.3	25.2	22.1	
Denis Goloubev	2	Counter	22	4	3.89	2	1.99	60	60	56.4	58.4	19.2	27.1	23.1	
Denis Goloubev	2	Counter	22	4	3.91	1	1	60	59.7	57.1	58.7	19.2	30	25.3	
Denis Goloubev	2	Parallel	22.6	4	3.88	4	3.99	60	59.3	54.5	57.1	19.3	24.4	22	
Denis Goloubev	2	Parallel	22.6	4	3.95	3	2.95	60	59.5	55.5	58	20.4	26.6	23.7	
Denis Goloubev	2	Parallel	22.6	4	3.89	2	2.02	60	59.4	55.7	57.9	19.2	27.4	23.6	
Denis Goloubev	2	Parallel	22.6	4	3.91	1	0.96	60	60.7	57.9	59.5	19.4	31.8	26.4	
Anh Hoang	1	Parallel	22.5	4	3.95	3	2.95	30	30.1	29	29.7	18.3	20	19.1	
Anh Hoang	1	Parallel	22.5	4	3.95	3	2.95	40	40.2	38.2	39.6	18.3	21.5	20.1	
Anh Hoang	1	Parallel	22.5	4	3.95	3	2.95	50	50.5	47.5	49.5	18.6	23.5	21.3	
Anh Hoang	1	Parallel	22.5	4	3.95	3	2.95	60	59.6	55.5	58	20.3	26.6	23.7	
Anh Hoang	1	Counter	21.7	4	3.98	3	2.97	30	30.3	29.1	29.9	18.3	20.1	19.1	
Anh Hoang	1	Counter	21.7	4	3.98	3	2.97	40	38.8	36.7	37.9	18.3	21.2	19.8	
Anh Hoang	1	Counter	21.7	4	3.98	3	2.97	50	48.7	45.5	47.7	18.4	22.9	20.6	
Anh Hoang	1	Counter	21.7	4	3.98	3	2.97	60	59.5	55.2	57.6	18.4	24.7	21.5	
Anh Hoang	2	Counter	22	4	3.9	4	4.02	60	59.4	54.6	57.2	18.4	23.5	20.8	
Anh Hoang	2	Counter	22	4	3.9	3	2.96	60	59.5	55.2	57.6	18.4	24.7	21.5	
Anh Hoang	2	Counter	22	4	3.89	2	1.98	60	59.1	55.6	57.7	18.4	26.4	22.4	
Anh Hoang	2	Counter	22	4	3.91	1	0.99	60	59.3	56	58	18.4	29.6	24.6	
Anh Hoang	2	Parallel	22.6	4	3.88	4	3.99	60	59.7	54.8	57.5	18.3	23.6	21.2	
Anh Hoang	2	Parallel	22.6	4	3.95	3	2.95	60	59.6	55.5	58	20.3	26.6	23.7	
Anh Hoang	2	Parallel	22.6	4	3.89	2	2.02	60	59.5	55.7	57.9	18.4	26.5	22.8	
Anh Hoang	2	Parallel	22.6	4	3.9	1	1.01	60	59.5	56.8	58.3	18.6	30.4	25.3	
Ngo Pham	1	Parallel	22.5	4	3.98	3	2.99	30	29.7	28.6	29.3	17.9	19.6	18.7	
Ngo Pham	1	Parallel	22.5	4	3.99	3	3.00	40	40.0	38.0	39.3	17.9	21.0	19.6	
Ngo Pham	1	Parallel	22.5	4	3.97	3	2.99	50	49.7	46.6	48.6	17.9	22.7	20.6	
Ngo Pham	1	Parallel	22.5	4	3.98	3	3.00	60	59.7	55.3	58.0	18.6	24.9	22.0	
Ngo Pham	1	Counter	21.7	4	3.98	3	3.00	30	29.8	28.8	29.4	18.0	19.7	18.8	
Ngo Pham	1	Counter	21.7	4	3.98	3	2.98	40	39.8	37.6	38.9	18.1	21.2	19.6	
Ngo Pham	1	Counter	21.7	4	3.99	3	2.96	50	49.6	46.3	48.1	18.2	22.8	20.4	
Ngo Pham	1	Counter	21.7	4	3.96	3	2.99	60	59.7	55.1	57.7	18.2	24.5	21.3	
Ngo Pham	2	Counter	22	4	3.86	4	4.00	60	59.0	54.1	56.8	18.2	22.3	20.6	
Ngo Pham	2	Counter	22	4	3.96	3	2.99	60	59.7	55.1	57.7	18.2	24.5	21.3	
Ngo Pham	2	Counter	22	4	3.86	2	2.00	60	59.8	56.1	58.3	18.3	26.5	22.4	
Ngo Pham	2	Counter	22	4	3.88	1	1.00	60	60.2	57.5	59.2	18.4	30.4	24.6	
Ngo Pham	2	Parallel	22.6	4	3.78	4	4.00	60	59.0	54.2	56.9	18.5	23.7	21.3	
Ngo Pham	2	Parallel	22.6	4	3.98	3	3.00	60	59.7	55.3	58.0	18.6	24.9	22.0	
Ngo Pham	2	Parallel	22.6	4	3.89	2	1.98	60	59.4	55.6	57.8	18.6	26.8	23.0	
Ngo Pham	2	Parallel	22.6	4	3.82	1	0.96	60	60.4	57.7	59.3	18.5	31.1	25.6	
Tran Dang	1	Parallel	22.5	4	3.95	3	2.95	30	29.8	28.8	29.4	18.3	19.9	19.1	
Tran Dang	1	Parallel	22.5	4	3.95	3	2.95	40	40.6	38.4	39.6	18.3	21.6	19.9	
Tran Dang	1	Parallel	22.5	4	3.95	3	2.95	50	49.5	46.6	48.5	18.5	23.3	21.1	
Tran Dang	1	Parallel	22.5	4	3.95	3	2.95	60	59.1	55	57.5	20.1	26.2	23.4	
Tran Dang	1	Counter	21.7	4	3.98	3	2.97	30	30.1	29	29.8	18.3	20	19.1	
Tran Dang	1	Counter	21.7	4	3.98	3	2.97	40	40.4	38.1	39.4	18.3	21.5	19.8	
Tran Dang	1	Counter	21.7	4	3.98	3	2.97	50	49.5	46.2	48	18.4	23	20.7	
Tran Dang	1	Counter	21.7	4	3.98	3	2.97	60	59.4	55.1	57.5	18.4	24.7	21.5	
Tran Dang	2	Counter	22	4	3.9	4	4.02	60	59.2	54.4	57	19.1	24.3	21.6	
Tran Dang	2	Counter	22	4	3.9	3	2.96	60	59.2	55	57.3	19.3	25.2	22.1	
Tran Dang	2	Counter	22	4	3.89	2	1.98	60	60	56.4	58.4	19.2	27.1	23.1	
Tran Dang	2	Counter	22	4	3.91	1	0.99	60	59.7	57.1	58.7	19.2	30	25.3	
Tran Dang	2	Parallel	22.6	4	3.88	4	3.99	60	59.3	54.5	57.1	19.3	24.4	22	
Tran Dang	2	Parallel	22.6	4	3.95	3	2.95	60	59.5	55.5	58	20.4	26.6	23.7	
Tran Dang	2	Parallel	22.6	4	3.89	2	2.02	60	59.4	55.7	57.9	19.2	27.4	23.6	
Tran Dang	2	Parallel	22.6	4	3.91	1	0.96	60	60.7	57.9	59.5	19.4	31.8	26.4	

Appendix 3 – Summarised Results

Summarised Pivot Table Experiment 1 Data									
Parallel/Counter Flow?	Counter								
Hot Flow Rate	4								
Responsibility	(All)								
Experiment Number	1								
Cold Flow Rate	(All)								
Row Labels	Average of TH1	Average of TH2	Average of TH3	Average of TC1	Average of TC2	Average of TC3	Var of Ambient Temperature	StdDev of Actual Cold Flow Rate	StdDev of Actual Hot Flow Rate
30	30.075	29	29.725	18.425	20.125	19.225	0	0.015	0
40	39.45	37.325	38.55	18.45	21.45	19.925	0	0.005	0
50	49.425	46.2	48.075	18.55	23.125	20.75	0	0.005	0.005
60	59.45	55.1	57.525	18.575	24.775	21.6	0	0.01	0.01
Grand Total	44.6	41.90625	43.46875	18.5	22.36875	20.375	4.85064E-13	0.00946847	0.005737305
Parallel/Counter Flow?	Parallel								
Hot Flow Rate	4								
Responsibility	(All)								
Experiment Number	1								
Cold Flow Rate	(All)								
Row Labels	Average of TH1	Average of TH2	Average of TH3	Average of TC1	Average of TC2	Average of TC3	Var of Ambient Temperature	StdDev of Actual Cold Flow Rate	StdDev of Actual Hot Flow Rate
30	29.85	28.8	29.475	18.4	20.05	19.2	0	0.02	0.015
40	40.15	38.15	39.475	18.525	21.7	20.2	0	0.025	0.02
50	50	46.975	48.95	18.7	23.5	21.325	0	0.02	0.01
60	59.475	55.325	57.875	19.85	26.075	23.2	0	0.025	0.015
Grand Total	44.86875	42.3125	43.94375	18.86875	22.83125	20.98125	0	0.02028957	0.01390436
Summarised Pivot Table Experiment 2 Data									
Parallel/Counter Flow?	Parallel								
Hot Flow Rate	4								
Responsibility	(All)								
Experiment Number	2								
Heater Temperature	(All)								
Row Labels	Average of TH1	Average of TH2	Average of TH3	Average of TC1	Average of TC2	Average of TC3	Var of Ambient Temperature	StdDev of Actual Cold Flow Rate	StdDev of Actual Hot Flow Rate
1	60.325	57.575	59.15	18.975	31.275	25.925	0	0.028867513	0.041932485
2	59.425	55.675	57.875	18.85	27.025	23.25	0	0.02	0
3	59.575	55.45	58	19.925	26.175	23.275	0	0.025	0.015
4	59.325	54.5	57.15	18.85	24.025	21.625	0	0.005	0.05
Grand Total	59.6625	55.8	58.04375	19.15	27.125	23.51875	0	1.152108213	0.049108723
Parallel/Counter Flow?	Counter								
Hot Flow Rate	4								
Responsibility	(All)								
Experiment Number	2								
Heater Temperature	(All)								
Row Labels	Average of TH1	Average of TH2	Average of TH3	Average of TC1	Average of TC2	Average of TC3	Var of Ambient Temperature	StdDev of Actual Cold Flow Rate	StdDev of Actual Hot Flow Rate
1	59.725	56.925	58.65	18.8	30	24.95	0	0.005773503	0.015
2	59.725	56.125	58.2	18.775	26.775	22.75	0	0.009574271	0.015
3	59.4	55.075	57.475	18.8	24.9	21.75	0	0.015	0.03
4	59.2	54.375	57	18.7	23.6	21.15	0	0.012583057	0.02
Grand Total	59.5125	55.625	57.83125	18.76875	26.31875	22.65	0	1.160363269	0.022656861





Appendix 4 – 3D Printed Model

